Visual Product Identity: Understanding identity perceptions conveyed by visual product design

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Abstract
This paper presents a study of the visual product identity of selected models of the Saab Automobile range of production and concept cars. In the study, the identity of the company's products as experienced internally within the company and externally by target customers was studied and compared. The purpose of the study was twofold; to understand and evaluate external and internal respondents' perception and interpretation of identity as conveyed by the visual design of the company's products, and to analyze the identity using VPI, an emerging model of visual product identity. In the VPI model, identity is defined by the three modes of recognition, comprehension and association, which refer to the different identity objects type, characteristics, and values, respectively. Results from the study show both similarities and differences in identity perceptions between internal and external respondents. Specifically, identity perceptions are rated higher among internal than among external respondents. Analysis suggests that there is an inherent hierarchical order between the modes with respect to their influence on identification, where the associative mode seems to have the greatest impact, followed by recognition and comprehension in order of importance. As different modes provide different perceptions of identity and also seem to be related to the complexity of the visual product form, this finding has important implications for product design.

Keywords
Visual product identity, VPI, design, experience, identification, brand, perception, recognition, comprehension, interpretation, association, semantics

Introduction

Companies need to offer products that display a strong and clear identity that is consistent, clear and appropriate as perceived by the target customer group. In saturated, mature market segments the role of the product form is crucial for identification in terms of creating brand awareness, product differentiation and communication of strategic brand messages.

In the VPI model (Visual Product Identity), the experience of identity based on the perception of visual references embodied in the product form is seen as a phenomenon which can be described by three modes of identification; recognition, comprehension and association. In VPI, the recognition mode includes the identification of product type through visual similarity with instances previously experienced. The comprehension mode refers to the understanding of product specific characteristics as presented by visual form elements. Through interpretation, the association mode provides value connotations through the creation of references known from previous experience and strategic brand communication.

When experiencing the product visually, perceivers are regarded as creating notions of product identity which can be mapped against these three modes. Through the three identification modes, the VPI model provides a full understanding of the different types of identity references that may be conveyed by visual product form.

What is identity?

Definitions of the concept of identity abound, each field with their own twitch to it. In the context of product design, we naturally understand the concept of identity in relation to products and to our relation with products.

As a phenomenon based on representation, we regard ‘identity’ as principally different from ‘aesthetics’, which is based on presentation (Vihma, 1995). Thus, we here understand aesthetics according to Baumgarten’s definition as ‘sensous delight’; the pleasure attained from sensory perception (Hekkert, 2006). The definition of identity as used here, implying that identification is dependent on meaning-making, means that rather than merely being perceived in the sense of awareness, identity must also be understood through signifying processes involving cognition – product identity is nothing without an understanding of the product, its context and its origin. This is part of the ‘anesthetic’ part of the product experience (Hekkert, 2006).

According to contemporary dictionaries, we can understand the concept of identity as “sameness of essential or generic character in different instances” or as “the distinguishing character or personality of an individual” (Merriam-Webster, 2006). If we transfer this definition to the domain of product design, we thus have to recognise two aspects of identity; on one hand, identity as an attribute of a thing, which is shared with something else (i.e., ‘similarity’); on the other hand, identity as a unique attribute of a thing (i.e., ‘dissimilarity’). The role of identification (i.e., the attribution of identity to a thing) through design is thus dual, a fact which is reflected in product design. On top of this, we also have to see identity in the social context of use; products are not only objects with meanings related to themselves, they are also critical for in socio-cultural communication and self identification. We use products to say something about ourselves; who we are, what we believe in, and what we value. Thus, identity conveyed by product design plays as much a role for producers

as it does for users, customers, owners, and the general public. Both producers and users communicate through the product; we are all part of an ongoing production and negotiation of identity messages through design.

In this research, identity is seen as a phenomenon involving signification; the creation of meanings based on semiotic signs. Meanings are constructed, maintained and evolved through representational qualities that are attributed to visual references of product form. Identification through design is regarded as a meaning-making practice that is carried by references within three domains of representation, referring to different types of 'identity objects', according to Table 1.

Identity object	Reference	Examples of meaning
Type	"What the product is"	function, use, purpose, brand (make)
Characteristics	"How the product is"	properties, performance, mode-of-use
Value	"What the product stands for"	origin, heritage, brand message

Table 1. Types of identity objects involved in identification through signification.

In semiotic terms, identity objects are the objects of signification; that, which the sign refers to. Thus, references in the mind of the perceiver connote a certain meaning, a certain understanding of the identity object. Ideally, but not necessarily, products provide us with all three types of references. Whether or not it has been explicitly considered during the design process, we are consciously and/or subconsciously informed about these forms of identity through references in the visual form of the product. Explicitly considering these three classes of identity objects, and understanding how they contribute to the creation of meaningful references through product form, may thus provide a significant advantage for the creation of desirable and meaningful products.

Method

In this section, the methods used to study identity creation based on visual product references are presented. Other methods were used to specifically study opinions, values and brand associations, see Fjellner and Stridsman-Dahlström (2006). In the research carried out for this paper, two principally different studies were carried out. First, the identity of the company's products as experienced by internal respondents within the company and by external respondents was studied in order to map identity references with visual elements of the studied models. Second, statements regarding expressive and formal attributes of the visual form of one specific model were compared. The objective was to study the consistency of specific expressive and descriptive meanings between internal and external respondents with respect to identity recognition and comprehension.

Internal study

The purpose of the internal study was to gain an understanding of how the brand and product strategy is perceived by internal respondents working on operative and strategic levels at Saab Automobile in Gothenburg. The objective was to reach insights into how identity strategy is practically implemented through product design references and how they are considered to support identification of Saab products.

In all, seven respondents within brand and design strategy, advanced design and marketing took part in the study. Through semi-structured interviews, topics covered included general issues surrounding the brand and values, its position on the market, the visual design of the cars, and identity references. Product image boards (see Figure 1) were used to provide a visual reference to the products discussed during the interviews. Production models studied in the research included 9-5 Sedan and Wagon, 9-3 Sport Sedan, Sport Wagon, and Convertible, 9-2X, and 9-7X. In order to study the relation between visual design elements and identity references, respondents also made a visual assessment of the 9X concept car. Image boards (Figure 2) were used as visual stimuli to assess the 9X model according to predefined terms in combination with visual analogue scales (Figure 3), see Wikström (2002). This part of the internal study served to provide material for the comparative study of internal and external respondents' identity perceptions.

External study

The purpose of the external study was to acquire an insight into external respondents' perceptions of the identity of Saab through studies of selected products. The objective of the study included mapping of identity references and visual elements. External respondents consisted of 44 university students in industrial design and industrial design engineering in Gothenburg, Delft, and London. In the study, these respondents were considered lead users representing the future target group of Saab in about ten years on the Swedish, Dutch, and English markets, respectively.

Partly based on results from the internal study, a semi-structured interview technique was used to cover a range of issues, including values in relation to cars, expressions related to the design of cars, and product related brand references. As for the internal study, a visual assessment of the 9X concept car was made using the same image boards (Figure 2) and visual analogue scales (Figure 3) to enable comparative analysis.

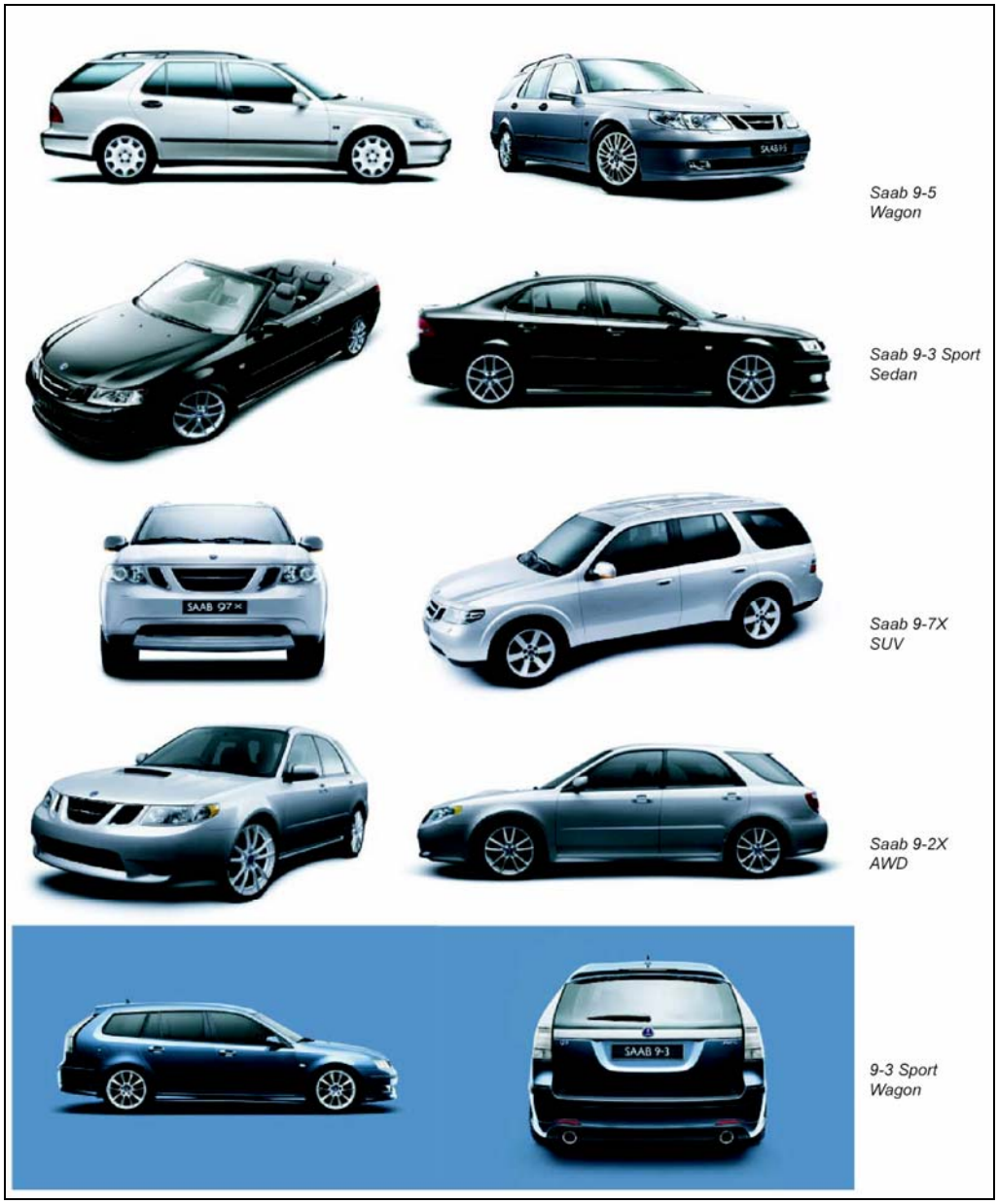


Figure 1. Product image board used as visual stimuli in the internal and external interview studies.



Figure 2. Image board of the 9X concept car used as visual stimuli in conjunction with visual analogue scales (see Figure 3) for comparative visual assessment during internal and external interview studies.

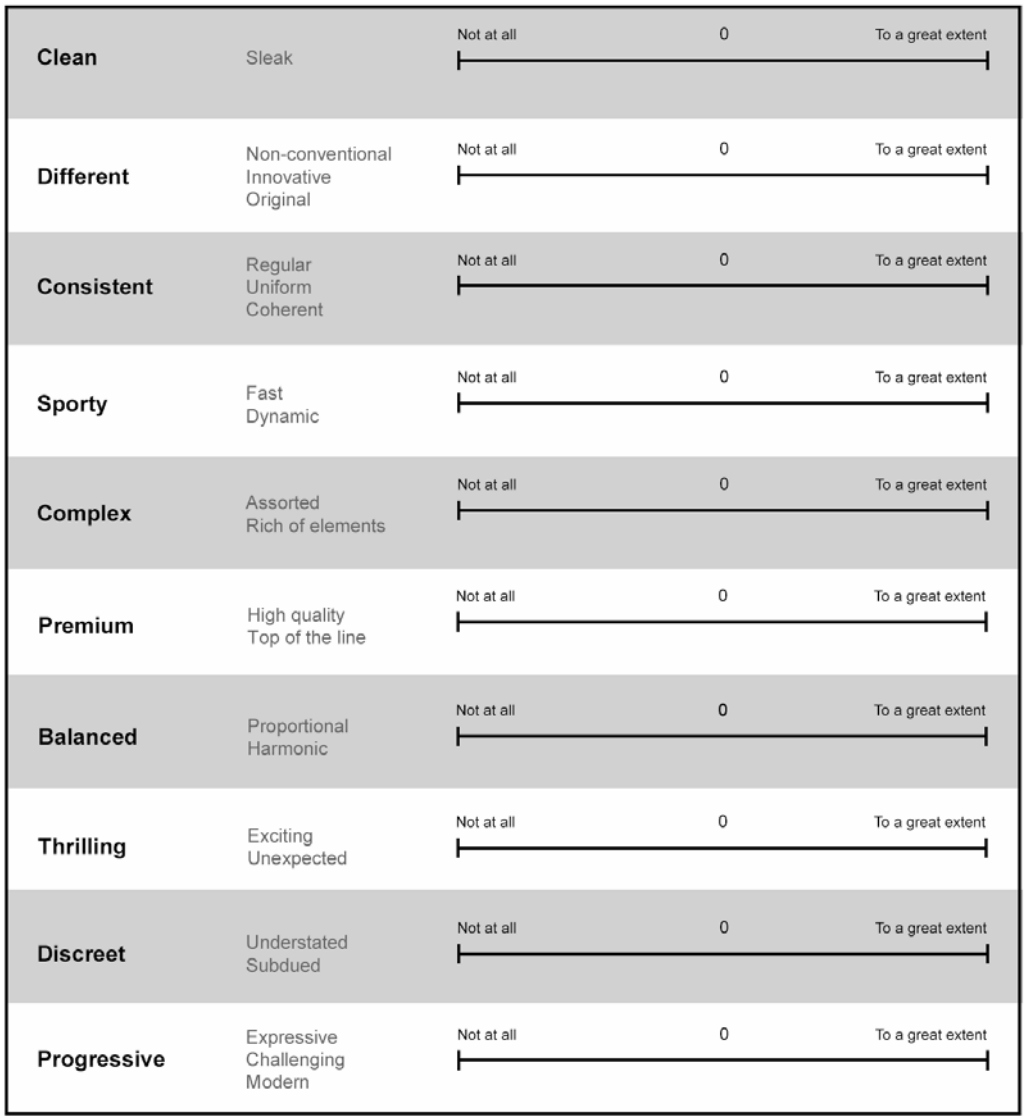


Figure 3. Visual analogue scales used for visual assessment during internal and external interview studies. Terms were chosen to allow an assessment of expressive and formal criteria in combination with image boards (see Figure 2).

Analysis framework: The VPI model

The model of Visual Product Identity (VPI) describes the possible representational modes that contribute to identification through visual design references and was used to analyse the findings from the studies. The VPI model (Figure 4) presents visual product identity as a meaning-making phenomenon that can be described by the three modes of identification, denoted ‘recognition’, ‘comprehension’, and ‘association’. These modes are related to the three types of semiotic signs – however, there is no absolute mapping between the type of sign and the form of identification. Rather, the modes refer to the type of meaning that is created in the mind of the perceiver, i.e. to which type of identity object (see Table 1) the sign refers to. This meaning is created through semiosis – the process of signifying (Chandler, 1994) – in the process of identification.

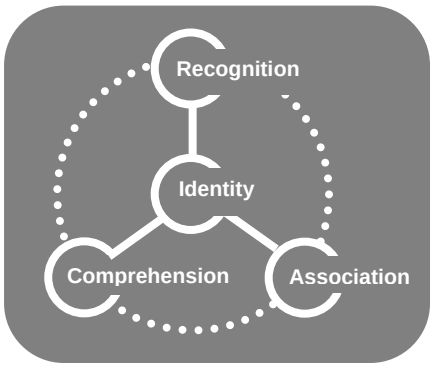


Figure 4. The model of Visual Product Identity (VPI), describing the modes of identification (meaning attribution) created through references denoted by the visual product form.

According to the model, perceivers are regarded as perceiving and attributing meaning to elements of the visual form of the product during the process of identification. These elements are denoted 'visual elements' and may be shapes, components, materials, colours, etc.; i.e. anything that is visually discernible to the viewer. Visual elements having any form meaning or representational quality to the viewer are denoted 'identifiers', i.e., they are active elements in the process of identification. In the mind of the viewer, identifiers *per se* function as references that denote certain meanings. From this follows that not all perceivers would necessarily consider the same visual elements as being identifiers, since not all visual elements would have meaning for all viewers. Depending on the nature of the perceiver, the identifier, and contextual factors, references convey meaning in one or several of the multiple VPI modes of recognition, comprehension, and association (see Table 2).

Element	>	Perception	>	Meaning
Identifier		Reference		Identification

Table 2. Schematic model of the process of identification. Identifiers (visual elements) of the product function as references during perception, which lead to meaningful identification in the modes of recognition, comprehension, and/or association. Visual elements that do not convey meaning would not be classified as identifiers.

Although each mode in the VPI model in the following is presented individually, they can never exist in isolation in the perception of product form, but are intertwined and impossible to distinguish at a phenomenological level.

The first mode, 'recognition', is based on identification through visual resemblance (similarity). Recognition thus requires previous visual precedents to compare with – identification through similarity is relative and thus depends on the existence of recognised likeness. Recognition, therefore, requires the presence of iconic signs, which function through a visual likeness between one thing (the icon) and another (the object). Iconic signs operate through motivation rather than interpretation (Joslyn, 1997) and their effect is immediate across cultures and customs (Wheeler

2003). Recognition can thus be considered the 'safest' form of identification as it provides a reliable way of communicating meaning in any market, provided it is consistently and clearly used.

Recognition can encompass several levels of identification referring to meaning within the general identity object 'type' (see Table 1). For example, recognition of product purpose or use involves similarity on the holistic form level through prototypical features (Muller, 2001) of the basic product sign (Monö, 1997) – a pair of scissors are identified through the sharpness of the cutting edges, an iron is identified by the flat, polished metal base, the electrical cord and an appropriately placed handle. Similarly, recognition of product brand requires resemblance to other products of a certain brand through similar visual elements primarily on the elementary level, often known as 'design cues'. For example, recognition of an Apple computer or music player is often achieved through consistent use of colour schemes and logotype.

The second mode, 'comprehension', relates to the identification through visual references of product specific characteristics such as properties, performance and mode-of-use. Through comprehension, we understand characteristics such as level of quality and the nature of the product; the product describes its operation, expresses its properties, and exhorts certain types of action or even non-action; it informs and advises about itself. This is an area where product semantics has contributed to the understanding of design (see, e.g. Monö 1997, Butter and Krippendorff 1984).

This form of identity attribution may be achieved through two types of signs depending on the type of visual reference. *The first form* operates through a physical connection or a causal relation between the signifier and the object; as such it is an indexical sign. An example of this is design quality; the even gap and tight flush between the different parts of an automobile body indicates a level of quality, which is the result of a high precision manufacturing process. As for recognition, there is thus a causal relation between process and outcome – 'it is what it seems to be'; there is an iconic motivation (Joslyn, 1997). *The second form* of comprehension operates through symbolic signs and is thus dependent on interpretation (Chandler, 1994). In a symbolic sign, the relationship between the signifier and the meaning is completely arbitrary – i.e., it is subjective or at best dictated by social convention or habit. Decoding of symbols is done within a given frame of reference; thus, the agreed meaning may differ between different target groups. For example, while a big city jeep for some may connote a sense of security and empowerment, for others it may be a symbol for unsustainable use of resources.

The third mode, 'association', is about communication of identity related to, e.g., values, origin and heritage, and works solely through symbolic signs. This mode of identification is dependent on subjective and socio-culturally conditioned processes of coding, which determine how we associate visual references with meaning through symbolic signs. In associative identification, meaning is interpreted (encoded and decoded) from two perspectives; from the point of view of the manufacturer, who uses the product to build brand values and convey strategic brand messages; and from the point of view of the customer or user, who communicates personal values and preferences through the use or ownership of the product. Customers extensively choose products based on the values they represent; they use the image of the product

brand to reflect their personality. The meanings of these signs are established primarily through strategic communication channels serving the purpose of brand messages and establishing preferred interpretations within interpretative communities on the market (Chandler 1994). However, meanings may also be formed outside the control of the branding process, creating products connotations that conflict with desired image.

Results

In the following sections, the results from two studies are presented. The first study, which focused on creating a general mapping of references between specific identifiers and modes of identification for various models as perceived by external respondents, is illustrated by the findings from the sub-study of the 9-7X production model. The second study focused on establishing a comparison between the assessments of internal and external respondents with respect to expressive and formal attributes of the visual form of the 9X concept car. Due to the qualitative nature of the studies and the limited number of respondents, no statistic analysis has been done. Rather, the intention is to provide an insight into typical results and to illustrate general trends.

Mapping of references: 9-7X model
The overall impression from respondents is that the 9-7X model is not perceived as a typical Saab. The reason for that impression is found mainly in the fact that the form as a whole has few recognizable Saab features. Even though the front end is perceived as a Saab in the recognition mode;

> “the only recognizable feature from other Saabs is the grille”

the rest of the form provides few references for brand identification through recognition:

> ”I don’t recognize the 9-7X as a Saab car because it has such a high front”
> “not as streamlined as other models”
> “it lacks the sloping waist line”
> “edgy and boxy”
> “too large”

Although a few of the respondents also mention the C-pillar, rear window and long headlights as recognizable features, these were a minority.



Figure 5. The Saab 9-7X.

In the comprehension mode, the most common statements refer to the overall expression of the car. Most of the respondents described the form of the 9-7X as "round, smooth, and bulgy", which was generally perceived as giving it a favourable expression on an overall level;

> "Strong expression"
> "Stable and reliable"

Despite this, the sporty and dynamic expression usually attributed to Saab cars was perceived as lacking. The opinion that the 9-7X looks ordinary – "just like any car" – was dominating.

In the association mode, respondents do not regard the model as a Saab due to the fact that it is a SUV – a type of car that respondents feel is neither 'typically' Saab, nor compatible with internally desired values such as 'understated' and 'Scandinavian'. Statements such as:

> "SUVs are for someone trying to prove himself"
> "SUVs are monstrous, aggressive and overdone"

clearly show that the association mode does not support consistent identification. In this case, the recognition of the characteristic Saab grille theme is not sufficiently strong to overcome this association.

Comparative Study: 9X Concept Car
Apart from the general mapping of references described previously, a comparative study of the 9X concept car (Figure 2) was made. The objective of this study was to compare assessments from internal and external respondents with respect to expression and formal properties of the visual product form.

As mentioned previously, the study utilized image boards (Figure 2) and visual analogue scales (Figure 3). Ten terms were used for the assessment. To support the desired interpretation of each term, they were specified with the help of descriptors according to Table 3. The terms used for the assessment were defined in collaboration with Saab Automobile in order to capture desired connotations with respect to visual expression and formal criteria. To allow for the comparative analysis, the same visual assessment was carried out with internal as well as external respondents.

Term	Descriptor
Clean	*Sleek*
Different	*Non-conventional, Innovative, Original*
Consistent	*Regular, Uniform, Coherent*
Sporty	*Fast, Dynamic*
Complex	*Assorted, Rich of elements*
Premium	*High quality, Top of the line*
Balanced	*Proportional, Harmonic*
Thrilling	*Exciting, Unexpected*
Discrete	*Understated, Subdued*
Progressive	*Expressive, Challenging, Modern*

Table 3. Expressive and formal terms used in the visual assessment of the 9X model.

As evident in Figure 6, the findings show a high degree of consistency between the assessments of internal and external respondents. The results show that identification was successful with respect to a high degree of consistency between internal and external identity assessments. A general trend in the results is that the mean value of the assessments of external respondents is consistently lower than for external respondents. Furthermore, the findings suggest that the correlation between internal and external respondents was higher with respect to expression than for formal attributes; two of the terms describing visual consistency (consistent and balanced) showed a greater spread than any of the expressive attributes. It is interesting to note that the third formal term, complex, on the contrary was most consistently perceived of all visual attributes.

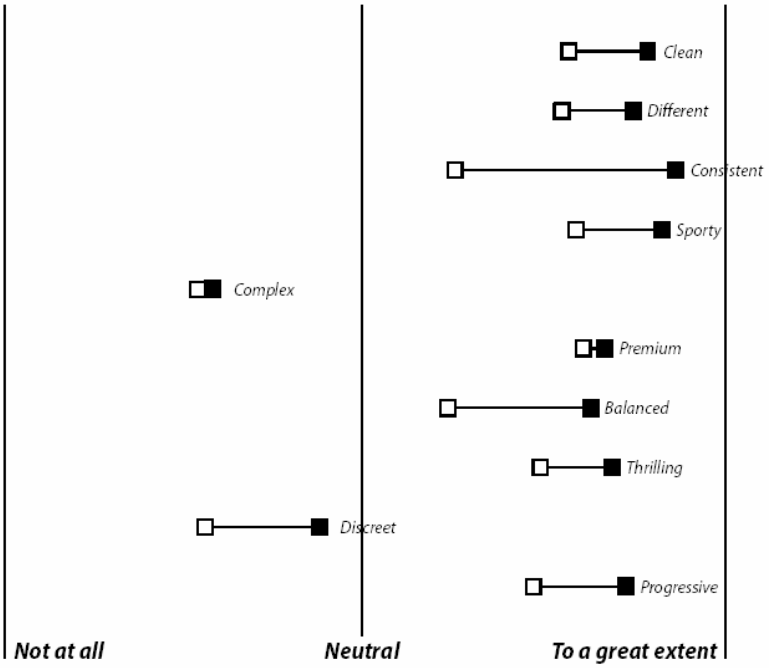


Figure 6. The visual assessment of the 9X concept car showed a great consistency between the 7 internal respondents (represented by black boxes) and the 44 external respondents (white boxes).

Discussion

Potentially, any visual element can function as an identifier, i.e., create meaningful references with respect to recognition, comprehension, and association. However, the studies show that certain identifiers (being either specific elements or compositions of elements) are more important for identification, i.e. they convey more meaning. Furthermore, the results suggest that there is also a hierarchical order between the modes of identification. Evidence suggests that the order of signification is related to the complexity of the identifiers: simple visual elements often connote less complex meaning (i.e. provide identification in lower identity modes such as recognition), while references that are more dependent on interpretation (such as comprehension and association) are connoted by compositions of visual elements. This understanding of the relation between modes of identification and complexity of visual elements corresponds with the multiple levels of product form proposed in work by Warell (2001). It is also supported by studies of Karjalainen and Warell (2004), showing that identification which is more dependent on interpretation requires more complex systems of identifiers, i.e. compositions of visual elements. In the following, findings related to the three modes will be discussed in further detail.

Identification through recognition

Mere recognition through visual elements seems to provide a rather weak form of identification. The studies suggest that recognition of brand, although important, is used primarily as the final piece of ‘evidence’ corroborating the identification already established through expression and association. If so, visual elements functioning only as recognition identifiers (and not in any other mode) would have a signifying function on par with logotypes or emblems, the main difference being that visual

elements legible from a greater distance also support visual differentiation. However, we also have keep in mind that recognition is supported by references identifying "what the product is", i.e. product purpose and use. Recognition as a mode of identification thus seems to work through signification from the least to the most complex levels of identifiers; i.e. from individual to compositions of visual elements.

Identification through association
The reasoning that more complex identifiers seem to dominate when it comes to identification is also supported by other findings in the study. External respondents tended to attribute their strongest identity references based on value connotations in the associative mode, where the dependence on composite visual elements is greater. In cases where the associative mode is stronger, it does not seem to matter whether there are in fact visual references in place to support identification through the recognition mode – the associative mode seems to have a greater impact on overall identification. This is exemplified by the studies of the 9-7X model – one of the strongest connotations was related to the perceived contradiction between the SUV type of vehicle and the values of Saab.

Identification through comprehension
If identification through recognition or association is not successful, respondents seem to create identification based on comprehension of the product itself. The comparative study of the 9X concept car supported the proposition that the product itself, through visual references expressing typical brand characteristics, can efficiently provide identification through comprehension – the strong consistency between internal and external assessments convincingly confirms this idea.

For products lacking strong identification in the recognition and association modes, identification through expressive visual elements may thus be a suitable strategy for identity attribution. In fact, this is often the only viable option for products and brands that are new on the market, and thus lack established brand values and recognisable elements. In such cases, perceivers are referred to creating identity references either through indexical references telling about product characteristics, or to symbolic references interpreted through previously established codes within the target group.

Multi-modal identification
A strong identifier has representation in more than one mode. This makes it more significant – i.e. it works as a sign in more than one way. From an identification perspective, visual elements that in this way work as multi-modal identifiers have several advantages: they have more meaning (will work as stronger signifiers), but since they are more complex in terms of visual composition, they are also more difficult to imitate. Identifiers that have meaning within one mode only run the risk of being perceived as non-genuine; a fact which is supported by the front-end design of the 9-7X, which, although featuring a recognizable grill, is still not perceived as authentic (i.e. coherent with brand values).

When multi-modal identifiers work well, they support each other and connote meaning in several modes. A typical example is the 'hockey stick'[1], which is

[1] A visual element in the form of an accelerated curve running from the rear side window along the waist line to the front end of the car, which is found in several contemporary and historic Saab models from the 1940s and onward.

perceived as a strong recognition identifier among internal as well as external respondents. Results from the studies indicate that apart from having significance with regard to recognition – this visual element it is perceived as a typical ‘Saab’ cue – as well as to comprehension – among external respondents, it is seen as expressing ‘sportiness’; among internal ‘speed’, ‘velocity’ and ‘dynamics’. Furthermore, it also contributes to associative identity: the expression ‘sportiness’ may be seen as a value which is associated with Saab’s heritage in rally sports and, more recently, with multisport sponsoring activities.

Conclusions

The study shows that visual product identity is a vast area of research where much awaits to be done. Still, it is evident from the studies made in this research that there is a great potential for understanding how producers and consumers create meaningful identity references through visual product form, and that significant differences between perceptions can be detected through such studies. For the practice of design the results have far-reaching implications, since they are attributable to specific visual elements and may thus potentially influence strategic design decisions.

Regarding the VPI model as such, it proved to be useful for categorizing statements by respondents and to create an understanding of the different ways that respondents create identity references through visual product form. Using the model as an analysis framework supported the formulation of principles for identity creation among respondents, and provided a structure for recommendations for future product identity development. Furthermore, the study methods and the VPI model were highly appreciated by the industrial partner as a potential tool for communication of identity related product design issues in internal processes.

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